

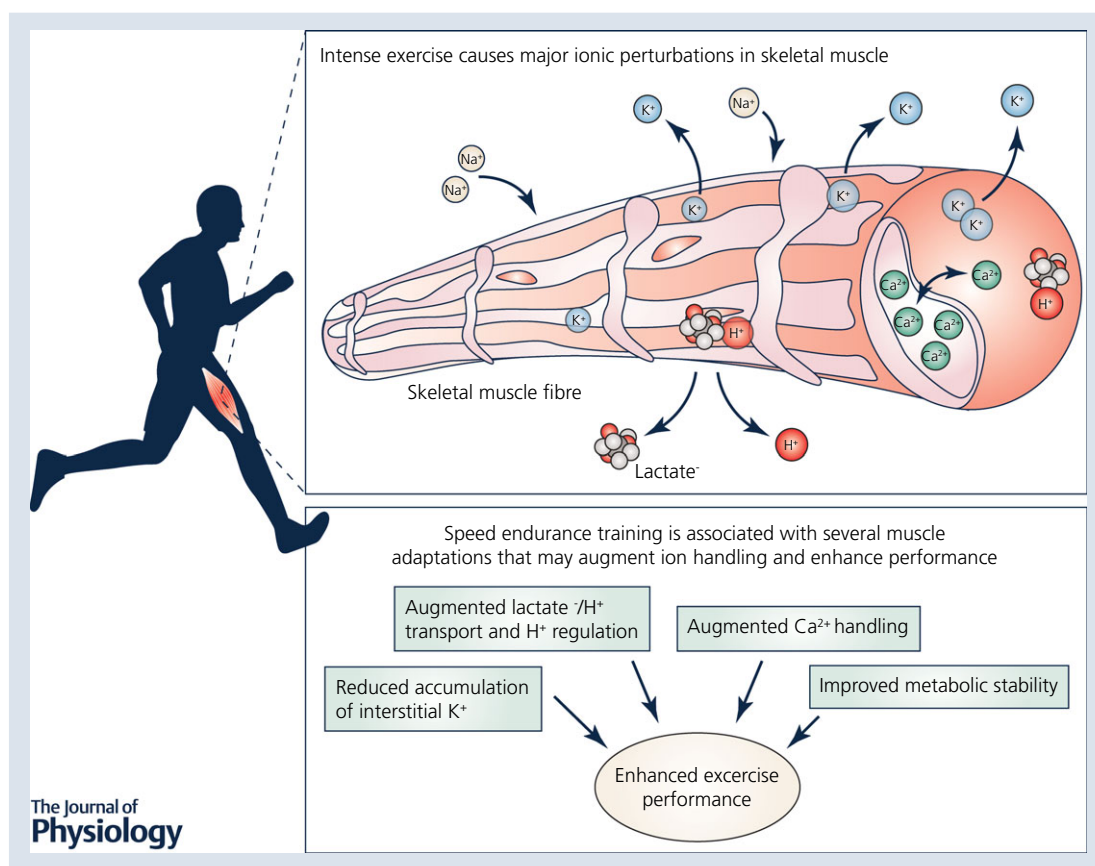
TOPICAL REVIEW

Limitations in intense exercise performance of athletes – effect of speed endurance training on ion handling and fatigue development

Morten Hostrup^{1,2} and Jens Bangsbo¹ 

¹Section of Integrated Physiology, Department of Nutrition, Exercise and Sports, University of Copenhagen, Denmark

²Department of Respiratory Research, Bispebjerg University Hospital, Denmark



Morten Hostrup, Ph.D., is Assistant Professor in the Department of Nutrition, Exercise and Sports (NEXS), University of Copenhagen and the Department of Respiratory Research at Bispebjerg Hospital in Copenhagen, Denmark. He studies fatigue processes and limitations in exercise performance of humans using various interventions, including training, nutrition and pharmacology. **Jens Bangsbo**, Ph.D., is Professor and Deputy Head at NEXS, University of Copenhagen, Denmark, and leader of the Integrative Physiology Group at NEXS. His research focuses on optimizing exercise performance, including how development of muscle fatigue is affected by high-intensity training.



Abstract Mechanisms underlying fatigue development and limitations for performance during intense exercise have been intensively studied during the past couple of decades. Fatigue development may involve several interacting factors and depends on type of exercise undertaken and training level of the individual. Intense exercise ($\frac{1}{2}$ –6 min) causes major ionic perturbations (Ca^{2+} , Cl^- , H^+ , K^+ , lactate $^-$ and Na^+) that may reduce sarcolemmal excitability, Ca^{2+} release and force production of skeletal muscle. Maintenance of ion homeostasis is thus essential to sustain force production and power output during intense exercise. Regular speed endurance training (SET), i.e. exercise performed at intensities above that corresponding to maximum oxygen consumption ($\dot{V}_{\text{O}_{2,\text{max}}}$), enhances intense exercise performance. However, most of the studies that have provided mechanistic insight into the beneficial effects of SET have been conducted in untrained and recreationally active individuals, making extrapolation towards athletes' performance difficult. Nevertheless, recent studies indicate that only a few weeks of SET enhances intense exercise performance in highly trained individuals. In these studies, the enhanced performance was not associated with changes in $\dot{V}_{\text{O}_{2,\text{max}}}$ and muscle oxidative capacity, but rather with adaptations in muscle ion handling, including lowered interstitial concentrations of K^+ during and in recovery from intense exercise, improved lactate $^-$ – H^+ transport and H^+ regulation, and enhanced Ca^{2+} release function. The purpose of this Topical Review is to provide an overview of the effect of SET and to discuss potential mechanisms underlying enhancements in performance induced by SET in already well-trained individuals with special emphasis on ion handling in skeletal muscle.

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Corresponding author M. Hostrup: Department of Nutrition, Exercise and Sports, University of Copenhagen, August Krogh, Universitetsparken 13, DK-2100, Denmark. Email: mhostrup@nexs.ku.dk

Abstract figure legend Intense exercise causes major ionic perturbations (Ca^{2+} , Cl^- , H^+ , K^+ , lactate $^-$ and Na^+) that may reduce sarcolemmal excitability, Ca^{2+} release, and force production of skeletal muscle. Maintenance of ion homeostasis is thus essential to sustain force production and power output during intense exercise. Speed endurance training, in which training sessions are performed at intensities above that corresponding to maximum oxygen consumption, is an efficient way to enhance intense exercise performance in already well-trained individuals. The enhancements in performance during intense exercise induced by speed endurance training may be related to muscle adaptations in ion handling, including improved regulation of K^+ homeostasis, enhanced lactate $^-$ – H^+ transport, and in some instances augmented H^+ buffering capacity, sarcoplasmic reticulum Ca^{2+} handling and glycolytic ATP provision.

Abbreviations CIC-1, Cl^- channel isoform 1; FXYD1, phospholemman; $\text{K}_{\text{IR}2.1}$, strong inward rectifying K^+ channel; K_{ATP} , ATP-sensitive K^+ channel; $\text{K}_{\text{IR}6.2}$, inward rectifying channel K^+ subunit 6.2; MCT1, monocarboxylate cotransporter isoform 1; MCT4, monocarboxylate cotransporter isoform 4; NHE1, Na^+ – H^+ exchanger isoform 1; NKCC1, Na^+ – K^+ – 2Cl^- exchanger isoform 1; ROS, reactive oxygen species; RyR1, ryanodine receptor isoform 1; SERCA1, sarcoplasmic reticulum Ca^{2+} ATPase isoform 1; SERCA2, sarcoplasmic reticulum Ca^{2+} ATPase isoform 2; SET, speed endurance training; SR, sarcoplasmic reticulum; $\dot{V}_{\text{O}_{2,\text{max}}}$, maximum oxygen consumption.

Introduction

Training is fundamental to improve exercise performance in the athlete. Optimization of training strategies for performance-enhancements is thus an important area of research within exercise physiology and sports medicine (Faria *et al.* 2005; Laursen, 2010; Bishop *et al.* 2011). In recent years, much research has focused on speed endurance training (SET), with brief intervals (≤ 60 s) performed at intensities above that corresponding to maximum oxygen consumption ($\dot{V}_{\text{O}_{2,\text{max}}}$) (Ross & Leveritt, 2001; Iaia & Bangsbo, 2010; Gibala & Jones, 2013; Bangsbo, 2015). As presented in Table 1, SET is an efficient way to enhance intense exercise performance ($\frac{1}{2}$ –6 min)

in already well-trained individuals and athletes, even with marked reductions in weekly training volume. It should be emphasised that in several of the studies presented in Table 1, training was also intensified by introducing or extending aerobic high-intensity training, which is characterized by interval training with intensities close to those eliciting $\dot{V}_{\text{O}_{2,\text{max}}}$.

The mechanisms underlying performance enhancements induced by a period of SET are not completely clear. Although several advanced human physiological experiments have been undertaken to investigate the effects of SET, a limitation of the research within this area is that most studies included untrained or recreationally active individuals (Iaia & Bangsbo, 2010; Gibala &

Table 1. Adaptations to SET in performance during intense exercise ($\frac{1}{2}$ –6 min) and skeletal muscle of trained individuals

Study	Subjects (no., M/F; exercise type)	$\dot{V}O_{2,max}$	Intervention (duration, sets/reps, work/rest, frequency, $\Delta\%$ volume)	Performance measure	Effect ($\Delta\%$)	Muscle adaptation	Effect ($\Delta\%$)
Houston & Thomson, 1977	5 M runners	59	6 wks of 3×60 s/120 s + 5×6 s/24 s + 2×90 s/180 s (4/wk) + leg press (15 RM)	Intense exercise (~ 1 min) 60 s running distance 90 s running distance	+17 +14 +13	ATP content	+15
Nevill et al. 1989	4 M/4F runners	55	8 wks of 2×30 s/10 min + $6 \times 10 \times 6$ s/54 s + $2 \times 5 \times 2$ min/5 min (3–4/wk)	30 s all out	+6	Rate of anaerobic glycolytic ATP production <i>In vivo</i> buffering capacity	+14 +44
Shepley et al. 1992	9 M runners	67	1 wk of $1 \times 5 \times \sim 70$ s/6–12 min (5/wk) –91% training volume	Intense exercise (~ 5 min)	+22	CS	+18
Dawson et al. 1998	9 M trained	57	6 wks of $3 \times 5 \times 4 \times 8 \times 10$ s/40–60 s (3/wk)	Repeated sprint Intense exercise (~ 1 min)	+2 +11	Phosphorylase activity CS activity	+40 –36
Ørtenblad et al. 2000	9 M trained	61	5 wks of 20×10 s/50 s (3/wk)	Repeated sprint	+12	Ca^{2+} release SERCAI SERCAII RyR1	+9 +41 +55 +48
Bickham et al. 2006	7 M runners	58	6 wks of $4 \times 14 \times 30 \times 40 \times 100$ m (3/wk)	Intense exercise ($\sim 2\frac{1}{2}$ min)	+11	Type II fibres MCT1	+6 +50
Bangsbo et al. 2009	12 M runners	63	6–9 wks. of $8 \times 12 \times 30$ s/180 s (2–3/wk) + 4×4 min (1–2/wk) –30%, training volume	Intense exercise (~ 2 min) 30 s all-out	+36 +5–7	$Na^+ - K^+$ subunit α_2 $Na^+ - K^+$ subunit β_1	+68 (+10)
Iaia et al. 2008, 2009	8/9 M runners	56	4 wks of $8 \times 12 \times 30$ s/180 s (3–4/wk) –64% training volume	Intermittent running Intense exercise ($\sim 1\frac{1}{2}$ min) 30 s all-out	+19 +19–27 +7	$Na^+ - K^+$ subunit α_1 $Na^+ - K^+$ subunit α_2 NKCC1 NHE1	+29 (+16) (+14) +30
Thomassen et al. 2010; Christensen et al. 2011	7 M football	55	2 wks of $10 \times 12 \times 30$ s/180 s (2/wk) –30% training volume	Repeated sprint Intermittent running	+2 (+6)	$Na^+ - K^+$ subunit α_2 FX/D1-phos MCT1 PDH	+15 +27 (+13) +17
Gunnarsson et al. 2012	7 M football	60	5 wks of $5 \times 9 \times 30$ s/180 s (1/wk) +11% training volume	Intermittent running	+11	MCT1 $Na^+ - K^+$ subunit β_1	+9 –13

(Continued)

Table 1. continued

Study	Subjects (no., M/F; exercise type)	$\dot{V}O_{2,max}$	Intervention (duration, sets/reps, work/rest, frequency, $\Delta\%$ volume)	Performance measure	Effect ($\Delta\%$)	Muscle adaptation	Effect ($\Delta\%$)
Gunnarsson <i>et al.</i> 2013; Thomassen <i>et al.</i> 2016	8 M cyclists	59	7 wks of 10–12 $\times$ 30 s/270 s (2–3/wk) + 4–5 $\times$ 3–4 min (1–2/wk) –70% training volume	Repeated sprint Intense exercise (~4 min)	+4 +18	<i>In vivo</i> buffering capacity CS COX4 $K_{IR2.1}$ $K_{IR6.2}$ NHE1 FXVD1 FXVD1-phos PLN PLN-phos CaMKII CaMKII-phos	+18 –16 –11 (+18) –14 (+30) +30 +40 +16 +42 +25 +300
Puype <i>et al.</i> 2013	9 M trained	53	6 wks of 4–9 $\times$ 30 s/270 s (3/wk)	–	–	PFK MCT1	+17 +70
Skovgaard <i>et al.</i> 2014	12 M runners	59	8 wks of 4–12 $\times$ 30 s/180 s + resistance training (4/wk) –42% training volume	1500 m running (~5 min) Intermittent running	+5 +44	NHE1 SERCAI	+35 –15
Nyberg <i>et al.</i> 2016	16 M football	–	9 wks of 2–3 $\times$ 8–10 $\times$ 5 s/10 s (1/2/wk)	Intermittent running	+12	HAD COX4 OXPHOS Capillarization	–8 –9 –16 –15
Vorup <i>et al.</i> 2016	9 M runners	60	8 wks of 4–8 $\times$ 30 s/180 s (2/wk) + resistance training/aerobic moderate or high intensity training (2/wk) –58% training volume	400 m running (~1 min) Intermittent running	+5 +19	Na^+K^+ subunit β_1 LDH	+15 +17

Relative effect size for performance and muscle changes from each study are only presented for statistical significant measures ($P \leq 0.05$) or when trending towards significance ($P = 0.05$ – 0.10) in parentheses. Abbreviations: CaMKII, Ca^{2+} –calmodulin-dependent protein kinase II; COX4, cytochrome c oxidase subunit 4; CS, citrate synthase; HAD, hydroxyacyl-CoA dehydrogenase; $K_{IR2.1}$, K^+ inward rectifier channel 2.1; $K_{IR6.2}$, K^+ inward rectifier channel 6.2; LDH, lactate dehydrogenase; MCT, monocarboxylate transporter; NHE1, Na^+H^+ exchanger isoform 1; NKCC1, $Na^+K^+2Cl^-$ channel isoform 1; OXPHOS, oxidative phosphorylation complex I–V enzymes; PFK, phosphofructokinase; PLN, phospholamban; RyR1, ryanodine receptor 1, SERCA, sarcoplasmic reticulum Ca^{2+} ATPase; $\dot{V}O_{2,max}$, maximum oxygen consumption ($ml\ kg^{-1}\ min^{-1}$).

Jones, 2013; Weston *et al.* 2014; Bangsbo, 2015), which makes translation towards athletes' performance difficult. In untrained and recreationally active individuals, SET augments $\dot{V}_{O_{2,max}}$ by 7–15% after only few weeks of training despite a low volume of training (Burgomaster *et al.* 2008; Jacobs *et al.* 2013). This observation is not surprising, since the heart rate of the individuals during a SET session with 6–10 $\times$ 30 s sprints, separated by 90 s of recovery, may reach mean and peaks of around 85 and 95% of maximum heart rate, respectively (Mohr *et al.* 2007). Furthermore, faster $\dot{V}_{O_2}$ kinetics, increased capillarization, and augmented mitochondrial function of skeletal muscle have been observed in untrained and recreationally active individuals following a period of SET (Jensen *et al.* 2004; Gibala *et al.* 2006; Jacobs *et al.* 2013; Christensen *et al.* 2016). In contrast, SET does not increase $\dot{V}_{O_{2,max}}$, capillarization or oxidative capacity of skeletal muscle in already trained individuals (Bangsbo *et al.* 2009; Iaia *et al.* 2009; Christensen *et al.* 2015). Actually, SET has been shown to enhance exercise performance in spite of a concurrent reduction in expression of oxidative enzymes and capillarization of skeletal muscle in well-trained individuals (Dawson *et al.* 1998; Nyberg *et al.* 2016). On the other hand, performance enhancements in trained individuals elicited by a period of SET are associated with several adaptations related to ion handling in skeletal muscle. SET may augment K^+ handling (Bangsbo *et al.* 2009; Thomassen *et al.* 2010, 2016), lactate $^-$ – H^+ transport capacity (Bickham *et al.* 2006; Gunnarsson *et al.* 2012, 2013; Puype *et al.* 2013), H^+ regulation (Nevill *et al.* 1989; Gunnarsson *et al.* 2013; Skovgaard *et al.* 2014), and Ca^{2+} handling function (Ørtenblad *et al.* 2000) of trained individuals. The importance of these adaptations for enhancements in performance during intense exercise ($\frac{1}{2}$ –6 min) following a period of SET in already trained individuals will be discussed in the present Topical Review.

Development of fatigue during intense exercise

Skeletal muscle fatigue can be defined as a disruption of the force production needed to meet the demand for a given exercise intensity (McKenna *et al.* 2008). The mechanisms underlying muscle fatigue development are highly debated and complex (Allen *et al.* 2008; Cairns & Lindinger, 2008; McKenna *et al.* 2008; MacIntosh & Shahi, 2011). Thus, fatigue development may involve several interacting factors and also depends on exercise modality and inter-individual differences (Knicker *et al.* 2011). Intense exercise ($\frac{1}{2}$ –6 min) causes major metabolic and ionic perturbations (Ca^{2+} , Cl^- , H^+ , K^+ , lactate $^-$, Mg^{2+} , Na^+ and P_i) that impair excitation–contraction coupling of skeletal muscle, decrease mechanical efficiency and lead to force decline (i.e. skeletal muscle fatigue) (Cairns & Lindinger, 2008; Cairns *et al.* 2015; Grassi

et al. 2015). Furthermore, accumulation of metabolites and ions in muscle interstitium stimulates sensory feedback from group III/IV muscle afferents to the central nervous system, thereby inducing the sensation of muscle ache associated with intense exercise (Pollak *et al.* 2014) and providing a regulatory input for motor command centres (Amann *et al.* 2011). Maintenance of ion homeostasis is thus essential to sustain force production and power output during intense exercise.

Skeletal muscle has several systems that regulate ion homeostasis of which some are presented in Fig. 1. The figure also provides information about the effect of a period of SET on adaptations in ion transporters, exchangers, and channels as well as in metabolic enzymes of trained individuals. The importance of these adaptations for performance during intense exercise will be discussed in the following sections.

Effect of speed endurance training on muscle K^+ homeostasis

Intense exercise causes major perturbations in the rundown of transmembrane ion gradients that may reduce sarcolemmal excitability and impair force production of skeletal muscle (Cairns & Lindinger, 2008; McKenna *et al.* 2008). Specifically, contraction-induced accumulation of extracellular K^+ depolarizes sarcolemma, which causes inexcitability and muscle force decline in isolated muscle preparations (Clausen, 2013, 2015; Cairns *et al.* 2015). During intense exercise in humans, the extracellular concentrations of K^+ rise from 4 mM to critical levels (> 10 mM), which, based on studies in muscle preparations, contribute to development of muscle fatigue (Cairns *et al.* 1995, 1997; Cairns & Lindinger, 2008; McKenna *et al.* 2008). Accordingly, concentrations of K^+ have been shown to exceed 12 mM in muscle interstitial fluid during intense one-legged knee-extensor exercise and cycling when measured with the microdialysis technique (Juel *et al.* 2000; Nielsen *et al.* 2003, 2004a; Nordsborg *et al.* 2003; Gunnarsson *et al.* 2013). Interstitial concentrations of K^+ are likely to be much higher than that observed because of the low time resolution, geometric placement of probes (e.g. outside the transverse-tubular system), and probe-to-probe variation associated with the microdialysis technique (Green *et al.* 2000; Juel *et al.* 2000; Nielsen *et al.* 2004a). In support of this hypothesis, extracellular concentrations of K^+ as high as 50–80 mM have been reported in rat extensor digitorum longus muscle following repeated stimulations, which were associated with pronounced muscle force decline (Clausen, 2008, 2011, 2013). K^+ handling is therefore of importance for maintenance of sarcolemmal excitability during muscle contractions and any system that counteracts accumulation of extracellular K^+ could potentially

delay fatigue development and enhance performance during intense exercise.

The Na^+/K^+ pump counteracts rundown of transmembrane gradients of Na^+ and K^+ , and is essential for muscle functioning during sustained contractions (Clausen, 2015). Regulation of Na^+/K^+ pump activity is multifactorial, with activity being stimulated by muscle contractions, intracellular Na^+ concentrations, and hormones, such as adrenaline and insulin (Clausen & Flatman, 1977; Clausen & Kohn, 1977; Flatman & Clausen, 1979; Clausen, 2003). Even a few days of training are capable of increasing the abundance of muscle Na^+/K^+ pumps by $\sim 15\%$ in recreationally active individuals (Green *et al.* 1993), and trained aged individuals have been shown to have a higher Na^+/K^+ pump abundance than untrained elderly subjects

(Klitgaard & Clausen, 1989). Intensified training has been shown to increase the abundance of muscle Na^+/K^+ pumps by $\sim 10\text{--}70\%$ (McKenna *et al.* 1993; Nielsen *et al.* 2004a; Bangsbo *et al.* 2009) when determined by both the $[^3\text{H}]$ -ouabain-binding technique (Madsen *et al.* 1994; Evertsen *et al.* 1997) and Western blotting (Iaia *et al.* 2008; Bangsbo *et al.* 2009; Thomassen *et al.* 2010; Vorup *et al.* 2016). Even in already well-trained individuals, a period of SET effectively increases expression of the catalytic α subunits and structural β subunits of the Na^+/K^+ pump. For instance, in trained runners, SET, with a concomitant reduction in training volume, has been shown to elevate expression of Na^+/K^+ pump subunits α_1 (29%), α_2 (15–68%) and β_1 (10–15%) (Iaia *et al.* 2008; Bangsbo *et al.* 2009). The increased abundance of muscle Na^+/K^+ pumps induced by SET is associated with a lower rate of

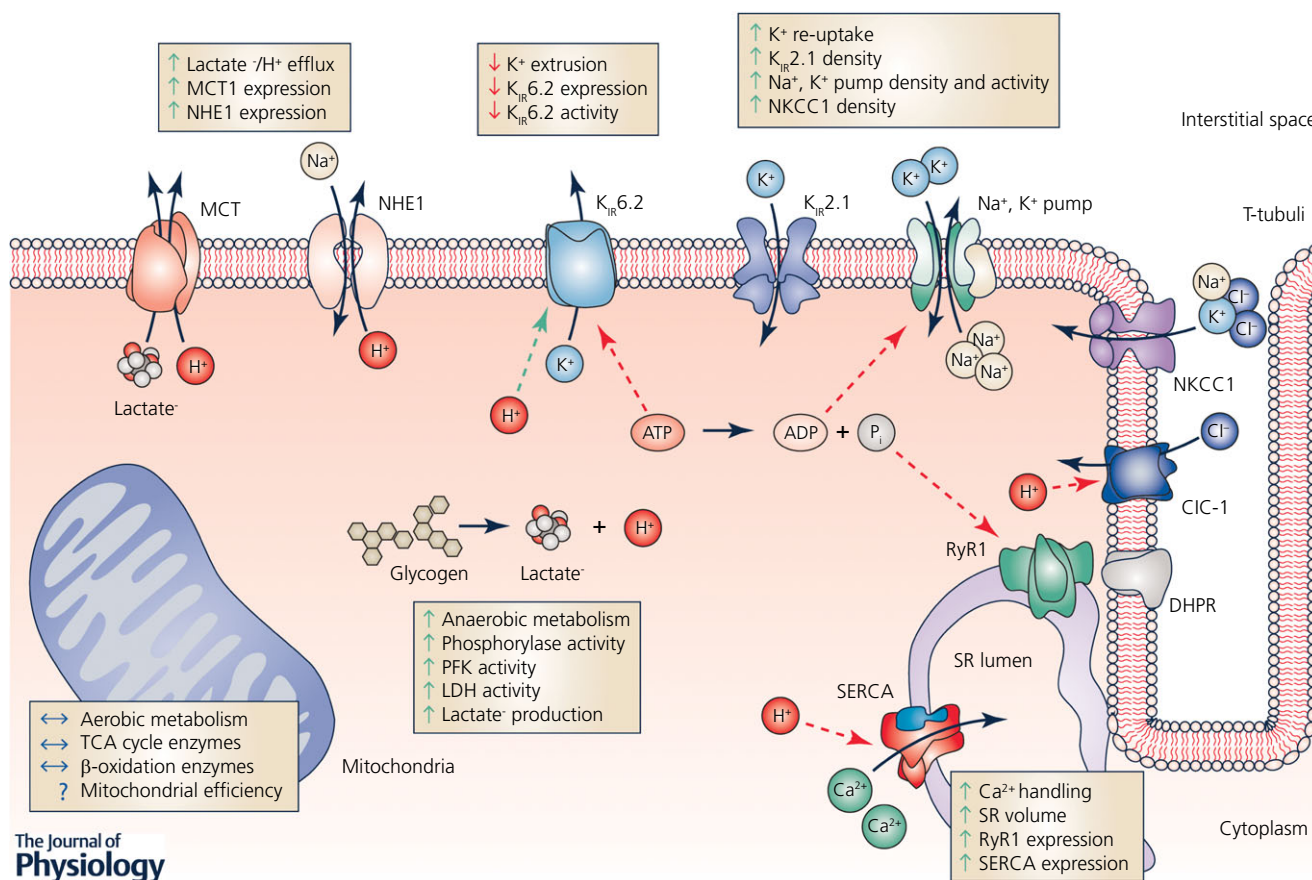


Figure 1. Overview of metabolic and ion handling systems in skeletal muscle of potential importance for performance during intense exercise, and their adaptations to a period of speed endurance training (SET)

Blue (continuous) arrows, direction of flux. Green (dashed) arrows, potentiation of a given protein/enzyme. Red (dashed) arrows, inhibition of a given protein/enzyme. Green (continuous) arrows, augmented by a period with SET. Red (continuous) arrows, lowered by a period with SET. Question marks (?), the effect of a period with SET is unknown. CIC-1, Cl⁻ channel isoform 1; DHPR, dihydropyridine receptor; K_{IR}2.1, K⁺ inward rectifier channel 2.1; K_{IR}6.2, K⁺ inward rectifier channel 6.2; LDH, lactate dehydrogenase; MCT, monocarboxylate transporter; NHE1, Na⁺-H⁺ exchanger isoform 1; NKCC1, Na⁺-K⁺-2Cl⁻ exchanger isoform 1; PFK, phosphofructokinase; RyR1, ryanodine receptor 1; SERCA, sarcoplasmic reticulum Ca²⁺ ATPase; SR, sarcoplasmic reticulum; TCA, tricarboxylic acid cycle.

K⁺ accumulation in arm venous blood during intense exercise and improved performance (Iaia *et al.* 2008; Bangsbo *et al.* 2009).

The question is, however, whether adaptations in muscle Na⁺–K⁺ pumps are essential for the improved performance observed following SET in trained individuals, since other studies did not find any change in Na⁺–K⁺ pump subunit abundances despite improved K⁺ handling (Gunnarsson *et al.* 2013) and performance during intense exercise (Gunnarsson *et al.* 2012, 2013; Skovgaard *et al.* 2014). In trained cyclists, Gunnarsson *et al.* (2013) observed that 7 weeks of intensified training consisting of SET and aerobic high-intensity training, with a 70% reduction in training volume, increased performance by 18% during intense exercise (~4 min) and reduced femoral venous plasma K⁺ concentrations by 0.7–0.8 mM in recovery from exercise, without alterations in expression of muscle Na⁺–K⁺ pump subunits. Thus, the lower femoral venous plasma K⁺ concentration in recovery from intense exercise following the intervention period may be related to other factors than changes in muscle Na⁺–K⁺ pump content. In support of this proposition, maximal *in vitro* Na⁺–K⁺ pump activity of muscle homogenate has been shown to be augmented with no contaminant change in abundance of Na⁺–K⁺ pump subunits in endurance athletes after 3 weeks of aerobic high intensity training (Aughey *et al.* 2007).

One such factor that regulates Na⁺–K⁺ pump activity and is induced by SET is phospholemman (FXYP1), which interacts with the α and β subunits of the Na⁺–K⁺ pump (Crambert *et al.* 2002; Thomassen *et al.* 2010, 2016). Phosphorylation of FXYP1 increases affinity for Na⁺ (lowers K_m) of the Na⁺–K⁺ pump and augments maximal ATPase activity ($\dot{V}_{max}$) (Crambert *et al.* 2002; Bibert *et al.* 2008; Rasmussen *et al.* 2008; Walas & Juel, 2012). In the aforementioned study in trained cyclists (Gunnarsson *et al.* 2013), expression of FXYP1 increased by 30% with the training period (Thomassen *et al.* 2016). In addition, non-specific FXYP1 phosphorylation was higher following the training at rest and during repeated intense exercise, mainly because of increased phosphorylation at serine68 (Thomassen *et al.* 2016), which regulates K_m of the Na⁺–K⁺ pump (Han *et al.* 2006; Juel *et al.* 2013). Similarly, 2 weeks of SET was shown to elevate phosphorylation of FXYP1 by 27% in elite football players, which correlated with enhancements in repeated sprint performance (10 × 20 m, 15 s active recovery) (Thomassen *et al.* 2010). Thus, SET-induced increases in expression and phosphorylation of FXYP1 may accentuate Na⁺–K⁺ pump activity and explain the lower femoral venous concentrations of K⁺ in recovery from intense exercise observed by Gunnarsson *et al.* (2013).

Aside from Na⁺–K⁺ pumps, skeletal muscle expresses Na⁺–K⁺–2Cl[–] exchangers (NKCC1), strong inward rectifying K⁺ channels (K_{IR}2.1), and ATP-sensitive K⁺

(K_{ATP}) channels, all of which are involved in K⁺ homeostasis of skeletal muscle and may be affected by SET (Spruce *et al.* 1985; Barrett-Jolley *et al.* 1999; Nielsen *et al.* 2003; Kristensen & Juel, 2010). However, only a few studies have investigated the effect of SET on muscle expression of NKCC1, K_{IR}2.1 and K_{ATP} channels in well-trained individuals. In trained runners, 4 weeks of SET was shown to increase expression of NKCC1 by 14% (Iaia *et al.* 2008). A higher expression of NKCC1 induced by SET may benefit ion homeostasis during intense exercise in two ways. Firstly, NKCC1 regulates myocellular volume via influx of K⁺ and Na⁺ with Cl[–] and water (Gosmanov *et al.* 2003, 2004; Cairns *et al.* 2015), thereby counteracting the osmotic changes associated with ionic perturbations during exercise. Secondly, NKCC1-mediated influx of Na⁺ stimulates Na⁺–K⁺ pump activity and thereby further re-uptake of K⁺ (Clausen & Nielsen, 2007). In addition, 7 weeks of SET have been shown to increase expression of K_{IR}2.1 by 18% ($P = 0.06$) and to reduce expression of the K_{ATP} channel subunit K_{IR}6.2 by 14% in trained cyclists (Gunnarsson *et al.* 2013). A higher expression of K_{IR}2.1 may augment K⁺ re-uptake when the electrochemical gradient for K⁺ is reduced because of interstitial K⁺ accumulation (Wallinga *et al.* 1999; Kristensen *et al.* 2006; Kristensen & Juel, 2010), whereas a lower density of K_{ATP} channel K_{IR}6.2 potentially reduces K⁺ extrusion to the interstitium during intense exercise, since critically low myocellular ATP levels cause opening of K_{IR}6.2 (Davies, 1990; Davies *et al.* 1992; Xu *et al.* 2001). The concurrent increase in expression of K_{IR}2.1 and reduction in expression of K_{IR}6.2 may therefore also explain the lower femoral venous plasma K⁺ observed in recovery from intense exercise and the improved performance observed by Gunnarsson *et al.* (2013). Nevertheless, the role of NKCC1, K_{IR}2.1 and K_{ATP} channels in K⁺ handling and performance during intense exercise is not completely clear. Studies using muscle preparations indicate the importance of these channels, since pharmacological manipulation of NKCC1, K_{IR}2.1 and K_{ATP} channels affects fatigue development (Gong *et al.* 2003; Kristensen *et al.* 2006; Selvin & Renaud, 2015). And although muscle fatigue processes in non-functioning K_{IR}6.2 mice models are uncertain (Boudreault *et al.* 2010; Bruton, 2010), K_{ATP} channels are highly expressed in the transverse-tubular system (Nielsen *et al.* 2003) and have been shown to serve a myoprotective role against high cytosolic Ca²⁺ fluctuations and fibre damage during intense muscle activity (Selvin & Renaud, 2015).

As outlined above, SET elicits several adaptations related to K⁺ handling of skeletal muscle in already trained individuals. SET may augment abundance and regulation of Na⁺–K⁺ pumps, increase expression of NKCC1 and K_{IR}2.1, and reduce expression of K_{IR}6.2. These adaptations are associated with lower concentrations of interstitial and femoral venous K⁺ during exercise and a more rapid

decline in recovery from exercise (Nielsen *et al.* 2004a; Gunnarsson *et al.* 2013). Given that the concentrations of interstitial and femoral venous K^+ reach same levels at exhaustion after compared to before a period of SET, the adaptations related to K^+ handling induced by SET may, at least in part, explain the improved performance during intense exercise in already trained individuals.

Effect of speed endurance training on accumulation of lactate⁻ and H^+

A major contributor to muscular energy production during intense exercise ($\frac{1}{2}$ –6 min) is glycolysis, leading to production of lactate⁻ and accumulation of H^+ (Hultman & Sahlin, 1980; Cheetham *et al.* 1986; Nevill *et al.* 1989; Gaitanos *et al.* 1993; Juel *et al.* 2004). Estimates of energy production derived from glycolysis range from 60–70% during 30 s of maximal cycling (Bogdanis *et al.* 1995; Kalsen *et al.* 2016) to 30–40% during 2–5 min of intense knee extensor exercise (Bangsbo *et al.* 1990, 1992). Thus, muscle lactate⁻ may accumulate from ~ 1 to > 25 mmol (kg ww)⁻¹ and pH may decline from ~ 7.2 to < 6.6 during intense exhaustive exercise lasting a few minutes (Juel *et al.* 1990; Bangsbo *et al.* 1993; Hostrup *et al.* 2014). Skeletal muscle has several transport systems that facilitate efflux of lactate⁻ and H^+ from myocellular compartments to the interstitium (Juel, 1998a; Juel & Halestrap, 1999; Pilegaard *et al.* 1999b). The main muscle transporters of lactate⁻ and H^+ during exercise are the lactate⁻– H^+ monocarboxylate cotransporters (MCT) of which the predominant isoforms 1 (MCT1) and 4 (MCT4) account for 70–80% of the lactate⁻ and H^+ efflux (Mason & Thomas, 1988; Juel, 1997; Pilegaard *et al.* 1999b; Thomas *et al.* 2012).

While several studies have investigated the effect of SET on muscle lactate⁻ and H^+ transport capacity and regulation in humans (Thomas *et al.* 2012), only few have been conducted in trained individuals. Unlike the marked improvements SET exerts in lactate⁻– H^+ transport capacity of skeletal muscle in untrained and recreationally active individuals (Pilegaard *et al.* 1999a; Juel *et al.* 2004; Burgomaster *et al.* 2007), the adaptive response to SET in trained individuals is moderate and sometimes not evident. A few weeks of SET with reduced training volume have been shown to increase expression of MCT1 by $\sim 10\%$ in elite football players and by up to $\sim 70\%$ in trained men (Bickham *et al.* 2006; Thomassen *et al.* 2010; Gunnarsson *et al.* 2012; Puype *et al.* 2013), whereas no changes were observed in trained runners and cyclists (Bangsbo *et al.* 2009; Gunnarsson *et al.* 2013; Vorup *et al.* 2016). Unlike MCT1, expression of MCT4 does not seem to change with a period of SET in trained individuals (Bickham *et al.* 2006; Thomassen *et al.* 2010; Gunnarsson *et al.* 2013; Puybe *et al.* 2013; Skovgaard *et al.* 2014). However, it may be that alterations in MCT4 only occur in sarcolemma,

since Dubouchaud *et al.* (2000) observed that endurance training increased expression of MCT4 in sarcolemmal fractions, but not in whole-muscle homogenate.

Apart from MCT, skeletal muscle has several systems that regulate myocellular H^+ , including Na^+ – H^+ exchangers (NHE1), carbonic anhydrase enzymes, HCO_3^- transporters, and H^+ buffers (Sahlin & Henriksson, 1984; Geers & Gros, 2000; Messonnier *et al.* 2007). While data on the effect of SET on carbonic anhydrase enzymes and HCO_3^- transporters are scant, studies have shown that SET increases content of NHE1 (Iaia *et al.* 2008; Skovgaard *et al.* 2014) and H^+ buffering capacity of skeletal muscle (Parkhouse & McKenzie, 1984; Bell & Wenger, 1988; Gibala *et al.* 2006; Baguet *et al.* 2011). NHE1 was shown to increase by 30–35% during a period of intensified training with reduced volume when SET was performed alone or in combination with resistance training in trained runners (Iaia *et al.* 2008; Skovgaard *et al.* 2014). Given that a higher expression of NHE1 may increase Na^+ uptake and H^+ efflux, it could be speculated that such adaptation may counteract contraction-induced accumulation of interstitial K^+ and sarcolemmal depolarization during intense exercise, since intracellular Na^+ stimulates Na^+ – K^+ pump activity (Nielsen *et al.* 2004b) and H^+ accumulation causes opening of K_{ATP} channels and hence extrusion of K^+ (Davies, 1990; Davies *et al.* 1992; Xu *et al.* 2001; Street *et al.* 2005). While most studies found no change in *in vitro* muscle H^+ buffering capacity of muscle homogenate in trained individuals during a period of SET (Nevill *et al.* 1989; Iaia *et al.* 2008), *in vivo* muscle buffering capacity has been shown to be increased in some studies (Nevill *et al.* 1989; Gunnarsson *et al.* 2013). Thus, Nevill *et al.* (1989) observed no change in buffering capacity of muscle homogenate *in vitro*, whereas the estimated *in vivo* muscle buffering capacity from H^+ and lactate⁻ during 30 s of maximal cycling was 44% higher after 8 weeks of SET in male and female runners. In line with this, Gunnarsson *et al.* (2013) observed that *in vivo* muscle buffering capacity estimated during intense exercise (~ 4 min) increased by $\sim 18\%$ with 8 weeks of combined SET and aerobic high-intensity training in trained cyclists. The difference between the *in vitro* and *in vivo* measurements observed may be due to a relative greater increase in the capacity of the trained muscle to release H^+ than lactate (Juel *et al.* 1998a,b), e.g. due to an increased expression of NHE1 (Gunnarsson *et al.* 2013), which *per se* will increase the estimated *in vivo* buffer capacity but not the *in vitro* equivalent.

The implications of adaptations in muscle lactate⁻– H^+ transport and H^+ regulation for performance during intense exercise following a period of SET in trained individuals are not completely clear. While SET increases rate of H^+ release during intense exercise and lactate⁻– H^+ clearance in the first few minutes of recovery from exercise, lactate⁻ and H^+ formation greatly exceed muscle

transport capacity during intense exercise (Bangsbo *et al.* 1990, 1993; Juel *et al.* 1990, 2004). To what extent accumulation of lactate⁻ and H⁺ contribute to fatigue development is also controversial (Korzeniewski & Zoladz, 2002; Pedersen *et al.* 2004; Bangsbo & Juel, 2006; Fitts, 2006; Lamb & Stephenson, 2006; Lindinger, 2007). Studies in muscle preparations have challenged the importance of lactate⁻ and H⁺ as muscle fatigue inducers (de Paoli *et al.* 2007), since low pH has little detrimental effect on fatigue development at physiological relevant temperatures, and lactate⁻ has been shown to restore the depressing effect of raised extracellular K⁺ on sarcolemmal excitability because of its inhibitory effect on Cl⁻ channels (ClC-1) (Nielsen *et al.* 2001; Pedersen *et al.* 2004; Hansen *et al.* 2005, 2005; de Paoli *et al.* 2010). In simulation models, Korzeniewski & Zoladz (2002) also demonstrated that H⁺ accumulation may counteract disturbances in energy system integrity, especially oxidative phosphorylation in working muscles, in that glycolytic production of H⁺ compensates the cytosolic alkalization caused by H⁺ consumption of creatine kinase at the onset of exercise. Nevertheless, although these studies in muscle preparations *in vitro* and in simulation models imply that lactate⁻ and H⁺ may even act protectively against fatigue (Allen & Westerblad, 2004), studies in humans indicate that accumulation of lactate⁻ and H⁺ may be detrimental for performance during intense exercise. For instance, accumulation of lactate⁻ induced by intense arm ergometer exercise was associated with greater accumulation of myocellular H⁺ (pH: 6.65 vs. 6.82), reduced lactate⁻ release (19 vs. 50 mmol), and a ~30% reduction in performance of the quadriceps during subsequent intense knee-extensor exercise (Bangsbo *et al.* 1996; Nordsborg *et al.* 2003). Notably, preceding arm ergometer exercise also caused a higher rate of extracellular K⁺ accumulation during the subsequent knee-extensor exercise (Bangsbo *et al.* 1996; Nordsborg *et al.* 2003). The latter observation may be related to the greater myocellular accumulation H⁺, thus causing opening of K_{ATP} channels (Davies, 1990; Davies *et al.* 1992; Xu *et al.* 2001; Street *et al.* 2005). In addition, lactate⁻-H⁺ transport and buffering capacity of skeletal muscle are good predictors for intense exercise performance (Nevill *et al.* 1989; Pilegaard *et al.* 1994; Bishop *et al.* 2003, 2004; Thomas *et al.* 2005; Messonnier *et al.* 2007), and athletes have higher lactate⁻-H⁺ transport and buffering capacity of skeletal muscle than trained and untrained individuals (McKenzie *et al.* 1982; Sahlin & Henriksson, 1984; Pilegaard *et al.* 1994; Edge *et al.* 2006). Lastly, β -alanine and sodium bicarbonate supplementation enhance intense exercise performance in trained individuals (Carr *et al.* 2011; Hobson *et al.* 2012). These effects are respectively related to increased muscle content of the H⁺ buffer carnosine (Hill *et al.* 2007) and systemic HCO₃⁻ concentrations (Sostaric *et al.* 2006). Augmented muscle

carnosine content may also increase Ca²⁺ sensitivity and affect cross-bridge formation because of carnosine's potential role as a diffusible Ca²⁺-H⁺ exchanger (Dutka *et al.* 2012; Swietach *et al.* 2013), whereas increased systemic HCO₃⁻ concentrations have been shown to reduce the rate of accumulation of extracellular H⁺ and K⁺ (Street *et al.* 2005; Sostaric *et al.* 2006). Collectively, these observations underpin the importance of improved muscle lactate⁻-H⁺ transport and H⁺ regulation induced by a period of SET for performance during intense exercise in already trained individuals.

Effect of speed endurance training on sarcoplasmic reticulum Ca²⁺ handling and myofibrillar Ca²⁺ sensitivity

Skeletal muscle force production and relaxation are regulated by myoplasmic concentrations of Ca²⁺ and myofibrillar Ca²⁺ sensitivity. Force production is controlled by Ca²⁺ release via opening of sarcoplasmic reticulum (SR) ryanodine receptors (RyR1) and relaxation by subsequent re-sequestration of Ca²⁺ from the myoplasm into the SR, which predominantly is controlled by the Ca²⁺ uptake rate of SR Ca²⁺-ATPase isoforms SERCAI and SERCAII, but also by the dissociation off-kinetics of troponin C, and the Ca²⁺ loading rate of Ca²⁺-binding proteins (Westerblad & Allen, 1996a,b; Nogueira *et al.* 2013). Based on studies in muscle preparations, impairment of Ca²⁺ release in particular appears to contribute to muscle fatigue development (Allen *et al.* 2008, 2011; Nogueira *et al.* 2013). However, while the importance of Ca²⁺ handling for fatigue resistance is well described in muscle preparations, surprisingly few studies have investigated the effect of training on SR Ca²⁺ handling function and myofibrillar Ca²⁺ sensitivity in humans, and only two studies have, to our knowledge, been undertaken with SET in trained individuals (Ørtenblad *et al.* 2000; Thomassen *et al.* 2016).

Ørtenblad *et al.* (2000) observed that 5 weeks of SET augmented maximal AgNO₃ stimulated Ca²⁺ release in SR vesicles of muscle homogenate by 9%, which was associated with a 12% better mean power output during a repeated sprint test (10 × 8 s, 32 s recovery) in highly trained men. The adaptations in Ca²⁺ release function were presumably related to a training-induced increase in total SR-volume, since expression of RyR1, SERCAI and SERCAII was increased by SET, whereas [H³]-ryanodine-specific binding remained unchanged (Ørtenblad *et al.* 2000). In spite of the increased expression of SERCAI and II, however, the authors observed no changes in SR vesicle Ca²⁺ uptake kinetics or Ca²⁺-ATPase activity of muscle homogenate with the intervention. Because these measures were performed in *in vitro* preparations, extrapolation to *in vivo* Ca²⁺ release and uptake function during exercise is speculative. However,

reduction in the *in vitro* SR Ca^{2+} release function of muscle homogenate has been shown to predict a decline in muscle torque after intense knee-extensor exercise of untrained individuals (Hill *et al.* 2001). In addition, 7 weeks of SET, combined with aerobic high-intensity training, was shown to increase total expression and level of phosphorylation of the SERCAII-regulatory protein phospholamban and Ca^{2+} -calmodulin-dependent protein kinase II (CaMKII) in cyclists (Thomassen *et al.* 2016). A higher expression and phosphorylation of phospholamban and CaMKII may increase the rate of SR Ca^{2+} release and re-uptake because of the reduced affinity for Ca^{2+} of SERCAII and increased opening probability of RyR1 (Suko *et al.* 1993; Rose *et al.* 2006; Kemi *et al.* 2007). Such effects might potentially improve SR Ca^{2+} handling during intense exercise. For myofibrillar Ca^{2+} sensitivity, currently no data are available regarding the effect of SET in trained individuals. In untrained individuals, Lynch *et al.* (1994) observed that sprint training lowered sensitivity for Sr^{2+} , while sensitivity for Ca^{2+} remained unaltered at normal pH but was lower at pH 6.6, which reflects the pH during intense exercise.

The limited number of studies that have investigated the effects of SET on Ca^{2+} handling makes interpretation of its implications difficult. Cross-sectional data have shown that endurance-trained individuals have lower *in vitro* Ca^{2+} handling function than resistance-trained and untrained individuals (Li *et al.* 2002). In line with this, aerobic training has been shown to reduce expression of SERCA and to lower *in vitro* Ca^{2+} release and uptake function of skeletal muscle (Majerczak *et al.* 2008; Green *et al.* 2011; Zoladz *et al.* 2013). Although these observations may imply that Ca^{2+} handling function is not the main determinant factor for endurance performance, studies have shown that Ca^{2+} uptake function may predict some of the variance during intense exercise (Harmer *et al.* 2014). Furthermore, 6 weeks of knee-extensor training was shown to increase rate of Ca^{2+} release and reduce Ca^{2+} leak from the SR, which were associated with enhanced peak power in untrained men (Munkvik *et al.* 2010). Nonetheless, while potential enhancing effects of SET in SR Ca^{2+} release function (Ørtenblad *et al.* 2000) as well as in expression and phosphorylation of phospholamban (Thomassen *et al.* 2016) may augment Ca^{2+} handling during intense exercise in trained individuals, more studies are needed on this issue.

Influence of speed endurance training on energy metabolism and reactive oxygen species: interaction with ion handling

Myocellular energy metabolism is important for excitation-contraction coupling during intense exercise. Na^+ - K^+ pumps and SERCA account for ~40% of energy expenditure of skeletal muscle during repeated

contractions (Clausen, 2003; Walsh *et al.* 2006; Barclay *et al.* 2007) and any local limitation in energy availability and/or accumulation of metabolites impairs ion handling and accelerates fatigue development (Matar *et al.* 2000; Gong *et al.* 2003; Nogueira *et al.* 2013; Ørtenblad *et al.* 2013; Place *et al.* 2015; Selvin & Renaud, 2015). Low levels of ATP with a concomitant rise in free Mg^{2+} , H^+ , ADP, IMP and P_i activates K_{ATP} channels (Allard *et al.* 1995; Barrett-Jolley & Davies, 1997; Pedersen *et al.* 2009), causes Ca^{2+} - P_i precipitation inside the SR (Allen *et al.* 2011), impairs Ca^{2+} handling (Li *et al.* 2002; Allen *et al.* 2008; Nogueira *et al.* 2013), and reduces Na^+ - K^+ pump activity (Juel *et al.* 2014). Furthermore, anaerobic energy production through glycolysis leads to lactate⁻ production and the accompanying H^+ accumulation changes SERCA activity and K_{ATP} channel sensitivity for ATP (Davies, 1990, 1992, Westerblad & Allen, 1993, Xu *et al.* 2001; Flagg *et al.* 2010; Nogueira *et al.* 2013). Anaerobic enzymes lactate dehydrogenase and creatine kinase may also interact with K_{ATP} channels (Crawford *et al.* 2002a,b), and Na^+ - K^+ pump activity and SR Ca^{2+} handling are regulated by glycogenolysis and glycolysis (James *et al.* 1999; Okamoto *et al.* 2001; Dutka & Lamb, 2007; Nogueira *et al.* 2013; Ørtenblad *et al.* 2013). As such, a higher capacity for glycolytic ATP provision following a period of SET may potentially improve regulation of ion homeostasis during intense exercise.

SET has been shown to accentuate glycolytic enzymes (Dawson *et al.* 1998; Iaia *et al.* 2008; Puype *et al.* 2013; Vorup *et al.* 2016), increase glycogen content, and augment capacity for glycolytic energy production of skeletal muscle in trained individuals (Nevill *et al.* 1989; Shepley *et al.* 1992). Furthermore, given that CaMKII regulates the rate of glycogenolysis and glycolysis (Rose *et al.* 2006), a higher expression and phosphorylation of CaMKII induced by SET may accelerate the rate of glycolytic ATP provision during intense exercise (Gunnarsson *et al.* 2013; Thomassen *et al.* 2016). Notably, Iaia *et al.* (2008) observed that glycogen and phosphocreatine utilization during a second bout of intense exercise to exhaustion (~1½ min) were lower after a period of SET, despite of higher accumulation of blood lactate⁻ and longer time to exhaustion. This observation suggests that muscle oxidation and/or mechanical efficiency was improved by SET. Such adaptations may enhance metabolic stability and delay the impairments in contraction-excitation contraction coupling of skeletal muscle during intense exercise (MacIntosh & Shahi, 2011; Grassi *et al.* 2015). Furthermore, any potential glycogen-sparing effect of a period of SET might potentially counteract exercise-induced reductions in SR Ca^{2+} handling, since critically low glycogen levels in proximity to the SR have been shown to be associated with impaired Ca^{2+} handling function in highly trained individuals (Ørtenblad *et al.* 2011; Hostrup *et al.* 2014; Nielsen *et al.* 2014).

Intense exercise leads to formation of reactive oxygen species (ROS) and glutathione in skeletal muscle that reacts with thiol groups (glutathionylation) of various myocellular proteins of importance for ion handling (Juel *et al.* 2015; Cheng *et al.* 2016). Among these proteins, the $\text{Na}^+ - \text{K}^+$ pump is subjected to glutathionylation in human skeletal muscle during intense exercise, which has been shown to attenuate maximal *in vitro* $\text{Na}^+ - \text{K}^+$ pump activity by $\sim 25\%$ in highly trained individuals (Hostrup *et al.* 2014; Juel *et al.* 2015); a phenomenon known as $\text{Na}^+ - \text{K}^+$ pump inactivation (McKenna *et al.* 2008). Furthermore, ROS-production reduces SR Ca^{2+} release function and myofibrillar Ca^{2+} sensitivity (Place *et al.* 2015; Cheng *et al.* 2016). Interestingly, in recreationally active individuals, muscle Ca^{2+} release channel RyR1 is subjected to fragmentation following one session of SET, whereas endurance athletes have no fragmentation (Place *et al.* 2015). This observation indicates that training augments muscle oxidant defence and protects the integrity of proteins related to ion handling during intense exercise. In support of these observations, a few weeks of SET has been shown to increase activity of anti-oxidant enzymes glutathione peroxidase and glutathione reductase in recreationally active individuals (Hellsten *et al.* 1996; Bogdanis *et al.* 2013). In addition, Green *et al.* (1998) observed that 7 and 11 weeks of high-intensity resistance training attenuated reductions in Ca^{2+} -ATPase activity induced by intense cycling. While it could be speculated that SET would have similar adaptive effects in already trained individuals, future studies are warranted to elucidate whether SET augments muscle oxidant defence and counteracts glutathionylation of the $\text{Na}^+ - \text{K}^+$ pump and fragmentation of RyR1 in the trained population.

Conclusion and perspectives

As presented in this review, only a few weeks of SET leads to significant performance enhancements during intense exercise in already trained individuals. Given that a period of SET has no apparent effects on $\dot{V}_{\text{O}_{2\text{max}}}$, haemoglobin mass, plasma volume, muscle mass, capillarization or oxidative capacity in trained individuals, the enhancements in performance during intense exercise induced by SET may be related to muscle adaptations in ion handling, including improved regulation of K^+ homeostasis, enhanced lactate $^- - \text{H}^+$ transport, and in some instances augmented H^+ buffering capacity and SR Ca^{2+} handling. The implications of these adaptations with relevance for performance during intense exercise may involve:

- Lower rate of accumulation of interstitial K^+ of exercising muscles due to increased $\text{Na}^+ - \text{K}^+$ pump activity and repression of K_{ATP} channels.
- Reduced extrusion of K^+ from K_{ATP} channels because of augmented lactate $^- - \text{H}^+$ transport and H^+ regulation.

- Augmented Ca^{2+} handling.
- Higher capacity for glycolytic ATP provision.

Because of the marked perturbations of multiple ions (Cairns & Lindinger, 2008; McKenna *et al.* 2008) and their interactions with energy production and ROS formation during muscle activity (MacIntosh & Shahi, 2011; Place *et al.* 2015; Cheng *et al.* 2016), fatigue processes during intense exercise in trained individuals are complex, and further studies are needed to elucidate the exact mechanisms underlying the enhancing effects of SET on performance. For instance, recent observations in muscle preparations indicate that extracellular Ca^{2+} interacts with lowered transsarcolemmal K^+ gradients during muscle contractions, where a reduction in extracellular Ca^{2+} lowers the intracellular activity of K^+ and contributes to fatigue development (Cairns *et al.* 2015). In this perspective, studies are warranted that investigate the importance of muscle $\text{Ca}^{2+} - \text{H}^+$ and $\text{Na}^+ - \text{Ca}^{2+}$ exchangers, as well as ClC-1 channels for exercise performance in humans and their potential adaptability to training. For example, ClC-1 channels are of interest because of their role in regulating Cl^- conductance and sarcolemmal excitability during muscle activity (Dutka *et al.* 2008; de Paoli *et al.* 2010, 2013). Furthermore, improvements related to ion handling, lactate $^-$ transport, and H^+ regulation after a period of SET could possibly affect regulation of myocellular and plasma osmolarity during exercise, and the implication of such changes for performance should be studied (Lindinger *et al.* 1995, 2013). Despite the difficulty in recruiting athletes for advanced physiological training interventions, more studies are needed in the athlete population. Lastly, it is relevant to probe the long-term effects of SET, since the intervention period of most studies that have provided an emphasis for this training modality only lasted a few weeks.

References

- Allard B, Fournet G, Rougier O, Descans B & Vivaudou M (1995). Dose-dependent activation and block by bisG10, a K^+ channel blocker, of mouse and frog skeletal muscle K_{ATP} channels. *FEBS Lett* **375**, 215–219.
- Allen D & Westerblad H (2004). Physiology. Lactic acid – the latest performance-enhancing drug. *Science* **305**, 1112–1113.
- Allen DG, Clugston E, Petersen Y, Röder IV, Chapman B & Rudolf R (2011). Interactions between intracellular calcium and phosphate in intact mouse muscle during fatigue. *J Appl Physiol* **111**, 358–366.
- Allen DG, Lamb GD & Westerblad H (2008). Skeletal muscle fatigue: cellular mechanisms. *Physiol Rev* **88**, 287–332.
- Amann M, Blain GM, Proctor LT, Sebranek JJ, Pegelow DF & Dempsey JA (2011). Implications of group III and IV muscle afferents for high-intensity endurance exercise performance in humans. *J Physiol* **589**, 5299–5309.

- Aughey RJ, Murphy KT, Clark SA, Garnham AP, Snow RJ, Cameron-Smith D, Hawley JA & McKenna MJ (2007). Muscle Na^+ - K^+ -ATPase activity and isoform adaptations to intense interval exercise and training in well-trained athletes. *J Appl Physiol* **103**, 39–47.
- Baguet A, Everaert I, De Naeyer H, Reyngoudt H, Stegen S, Beeckman S, Achten E, Vanhee L, Volkaert A, Petrovic M, Taes Y & Derave W (2011). Effects of sprint training combined with vegetarian or mixed diet on muscle carnosine content and buffering capacity. *Eur J Appl Physiol* **111**, 2571–2580.
- Bangsbo J & Juel C (2006). Counterpoint: lactic acid accumulation is a disadvantage during muscle activity. *J Appl Physiol* **100**, 1412–1413.
- Bangsbo J (2015). Performance in sports – With specific emphasis on the effect of intensified training. *Scand J Med Sci Sports* **25**, 88–99.
- Bangsbo J, Gollnick PD, Graham TE, Juel C, Kiens B, Mizuno M & Saltin B (1990). Anaerobic energy production and O_2 deficit-debt relationship during exhaustive exercise in humans. *J Physiol* **422**, 539–559.
- Bangsbo J, Graham T, Johansen L, Strange S, Christensen C & Saltin B (1992). Elevated muscle acidity and energy production during exhaustive exercise in humans. *Am J Physiol* **263**, 891–899.
- Bangsbo J, Gunnarsson TP, Wendell J, Nybo L & Thomassen M (2009). Reduced volume and increased training intensity elevate muscle Na^+ - K^+ pump $\alpha 2$ -subunit expression as well as short- and long-term work capacity in humans. *J Appl Physiol* **107**, 1771–1780.
- Bangsbo J, Johansen L, Graham T & Saltin B (1993). Lactate and H^+ effluxes from human skeletal muscles during intense, dynamic exercise. *J Physiol* **462**, 115–133.
- Bangsbo J, Madsen K, Kiens B & Richter EA (1996). Effect of muscle acidity on muscle metabolism and fatigue during intense exercise in man. *J Physiol* **495**, 587–596.
- Barclay CJ, Woledge RC & Curtin NA (2007). Energy turnover for Ca^{2+} cycling in skeletal muscle. *J Muscle Res Cell Motil* **28**, 259–274.
- Barrett-Jolley R & Davies NW (1997). Kinetic analysis of the inhibitory effect of glibenclamide on K_{ATP} channels of mammalian skeletal muscle. *J Membr Biol* **155**, 257–262.
- Barrett-Jolley R, Dart C & Standen NB (1999). Direct block of native and cloned (Kir2.1) inward rectifier K^+ channels by chloroethylclonidine. *Br J Pharmacol* **128**, 760–766.
- Bell GJ & Wenger HA (1988). The effect of one-legged sprint training on intramuscular pH and nonbicarbonate buffering capacity. *Eur J Appl Physiol Occup Physiol* **58**, 158–164.
- Bibert S, Roy S, Schaer D, Horisberger JD & Geering K (2008). Phosphorylation of phospholemman (FXD1) by protein kinases A and C modulates distinct Na,K -ATPase isozymes. *J Biol Chem* **283**, 476–486.
- Bickham DC, Bentley DJ, Le Rossignol PF & Cameron-Smith D (2006). The effects of short-term sprint training on MCT expression in moderately endurance-trained runners. *Eur J Appl Physiol* **96**, 636–643.
- Bishop D, Edge J & Goodman C (2004). Muscle buffer capacity and aerobic fitness are associated with repeated-sprint ability in women. *Eur J Appl Physiol* **92**, 540–547.
- Bishop D, Girard O & Mendez-Villanueva A (2011). Repeated-sprint ability – part II: recommendations for training. *Sports Med* **41**, 741–756.
- Bishop D, Lawrence S & Spencer M (2003). Predictors of repeated-sprint ability in elite female hockey players. *J Sci Med Sport* **6**, 199–209.
- Bogdanis GC, Nevill ME, Boobis LH, Lakomy HK & Nevill AM (1995). Recovery of power output and muscle metabolites following 30 s of maximal sprint cycling in man. *J Physiol* **482**, 467–480.
- Bogdanis GC, Stavrinou P, Fatouros IG, Philippou A, Chatzinikolaou A, Draganidis D, Ermidis G & Maridaki M (2013). Short-term high-intensity interval exercise training attenuates oxidative stress responses and improves antioxidant status in healthy humans. *Food Chem Toxicol* **61**, 171–177.
- Boudreault L, Cifelli C, Bourassa F, Scott K & Renaud JM (2010). Fatigue preconditioning increases fatigue resistance in mouse flexor digitorum brevis muscles with non-functioning K_{ATP} channels. *J Physiol* **588**, 4549–4562.
- Bruton J (2010). What does the membrane K_{ATP} channel really do in skeletal muscle? *J Physiol* **588**, 4333.
- Burgomaster KA, Cermak NM, Phillips SM, Benton CR, Bonen A & Gibala MJ (2007). Divergent response of metabolite transport proteins in human skeletal muscle after sprint interval training and detraining. *Am J Physiol Regul Integr Comp Physiol* **292**, R1970–R1976.
- Burgomaster KA, Howarth KR, Phillips SM, Rakobowchuk M, Macdonald MJ, McGee SL & Gibala MJ (2008). Similar metabolic adaptations during exercise after low volume sprint interval and traditional endurance training in humans. *J Physiol* **586**, 151–160.
- Cairns SP & Lindinger MI (2008). Do multiple ionic interactions contribute to skeletal muscle fatigue? *J Physiol* **586**, 4039–4054.
- Cairns SP, Flatman JA & Clausen T (1995). Relation between extracellular $[\text{K}^+]$, membrane potential and contraction in rat soleus muscle: modulation by the Na^+ - K^+ pump. *Pflügers Arch* **430**, 909–915.
- Cairns SP, Hing WA, Slack JR, Mills RG & Loiselle DS (1997). Different effects of raised $[\text{K}^+]_o$ on membrane potential and contraction in mouse fast- and slow-twitch muscle. *Am J Physiol* **273**, 598–611.
- Cairns SP, Leader JP, Loiselle DS, Higgins A, Lin W & Renaud JM (2015). Extracellular Ca^{2+} -induced force restoration in K^+ -depressed skeletal muscle of the mouse involves an elevation of $[\text{K}^+]_i$: implications for fatigue. *J Appl Physiol* **118**, 662–674.
- Carr AJ, Hopkins WG & Gore CJ (2011). Effects of acute alkalosis and acidosis on performance: a meta-analysis. *Sports Med* **41**, 801–814.
- Cheetham ME, Boobis LH, Brooks S & Williams C (1986). Human muscle metabolism during sprint running. *J Appl Physiol* **61**, 54–60.
- Cheng AJ, Yamada T, Rassier D, Andersson DC, Westerblad H & Lanner JT (2016). Reactive oxygen/nitrogen species and contractile function in skeletal muscle during fatigue and recovery. *J Physiol* **594**, 5149–5160.

- Christensen PM, Gunnarsson TP, Thomassen M, Wilkerson DP, Nielsen JJ, Bangsbo J (2015). Unchanged content of oxidative enzymes in fast-twitch muscle fibers and $\dot{V}O_2$ kinetics after intensified training in trained cyclists. *Physiol Rep* **3**, pii: e12428.
- Christensen PM, Jacobs RA, Bonne T, Flück D, Bangsbo J & Lundby C (2016). A short period of high-intensity interval training improves skeletal muscle mitochondrial function and pulmonary oxygen uptake kinetics. *J Appl Physiol* **120**, 1319–1327.
- Christensen PM, Krstrup P, Gunnarsson TP, Kiilerich K, Nybo L & Bangsbo J (2011). $\dot{V}O_2$ kinetics and performance in soccer players after intense training and inactivity. *Med Sci Sports Exerc* **43**, 1716–1724.
- Clausen T & Flatman JA (1977). The effect of catecholamines on Na-K transport and membrane potential in rat soleus muscle. *J Physiol* **270**, 383–414.
- Clausen T & Kohn PG (1977). The effect of insulin on the transport of sodium and potassium in rat soleus muscle. *J Physiol* **265**, 19–42.
- Clausen T & Nielsen OB (2007). Potassium, Na^+ , K^+ -pumps and fatigue in rat muscle. *J Physiol* **584**, 295–304.
- Clausen T (2003). Na^+ - K^+ pump regulation and skeletal muscle contractility. *Physiol Rev* **83**, 1269–1324.
- Clausen T (2008). Clearance of extracellular K^+ during muscle contraction – roles of membrane transport and diffusion. *J Gen Physiol* **131**, 473–481.
- Clausen T (2011). In isolated skeletal muscle, excitation may increase extracellular K^+ 10-fold; how can contractility be maintained? *Exp Physiol* **96**, 356–368.
- Clausen T (2013). Excitation-induced exchange of Na^+ , K^+ , and Cl^- in rat EDL muscle *in vitro* and *in vivo*: physiology and pathophysiology. *J Gen Physiol* **141**, 179–192.
- Clausen T (2015). Excitation of skeletal muscle is a self-limiting process, due to run-down of Na^+ , K^+ gradients, recoverable by stimulation of the Na^+ , K^+ pumps. *Physiol Rep* **3**, pii: e12373.
- Crambert G, Fuzesi M, Garty H, Karlish S & Geering K (2002). Phospholemman (FXD1) associates with Na,K-ATPase and regulates its transport properties. *Proc Natl Acad Sci USA* **99**, 11476–11481.
- Crawford RM, Budas GR, Jovanović S, Ranki HJ, Wilson TJ, Davies AM & Jovanović A (2002a). M-LDH serves as a sarcolemmal K_{ATP} channel subunit essential for cell protection against ischemia. *EMBO J* **21**, 3936–39348.
- Crawford RM, Ranki HJ, Botting CH, Budas GR & Jovanovic A (2002b). Creatine kinase is physically associated with the cardiac ATP-sensitive K^+ channel *in vivo*. *FASEB J* **16**, 102–104.
- Davies NW (1990). Modulation of ATP-sensitive K^+ channels in skeletal muscle by intracellular protons. *Nature* **343**, 375–377.
- Davies NW, Standen NB & Stanfield PR (1992). The effect of intracellular pH on ATP-dependent potassium channels of frog skeletal muscle. *J Physiol* **445**, 549–568.
- Dawson B, Fitzsimons M, Green S, Goodman C, Carey M & Cole K (1998). Changes in performance, muscle metabolites, enzymes and fibre types after short sprint training. *Eur J Appl Physiol Occup Physiol* **78**, 163–169.
- de Paoli FV, Broch-Lips M, Pedersen TH & Nielsen OB (2013). Relationship between membrane Cl^- conductance and contractile endurance in isolated rat muscles. *J Physiol* **591**, 531–545.
- de Paoli FV, Ørtenblad N, Pedersen TH, Jørgensen R & Nielsen OB (2010). Lactate *per se* improves the excitability of depolarized rat skeletal muscle by reducing the Cl^- conductance. *J Physiol* **588**, 4785–4794.
- de Paoli FV, Overgaard K, Pedersen TH & Nielsen OB (2007). Additive protective effects of the addition of lactic acid and adrenaline on excitability and force in isolated rat skeletal muscle depressed by elevated extracellular K^+ . *J Physiol* **581**, 829–839.
- Dubouchaud H, Butterfield GE, Wolfel EE, Bergman BC & Brooks GA (2000). Endurance training, expression, and physiology of LDH, MCT1, and MCT4 in human skeletal muscle. *Am J Physiol Endocrinol Metab* **278**, E571–E579.
- Dutka TL & Lamb GD (2007). Na^+ - K^+ pumps in the transverse tubular system of skeletal muscle fibers preferentially use ATP from glycolysis. *Am J Physiol Cell Physiol* **293**, C967–C977.
- Dutka TL, Lambole CR, McKenna MJ, Murphy RM & Lamb GD (2012). Effects of carnosine on contractile apparatus Ca^{2+} sensitivity and sarcoplasmic reticulum Ca^{2+} release in human skeletal muscle fibers. *J Appl Physiol* **112**, 728–736.
- Dutka TL, Murphy RM, Stephenson DG & Lamb GD (2008). Chloride conductance in the transverse tubular system of rat skeletal muscle fibres: importance in excitation–contraction coupling and fatigue. *J Physiol* **586**, 875–887.
- Edge J, Bishop D, Hill-Haas S, Dawson B & Goodman C (2006). Comparison of muscle buffer capacity and repeated-sprint ability of untrained, endurance-trained and team-sport athletes. *Eur J Appl Physiol* **96**, 225–234.
- Evertsen F, Medbo JJ, Jebens E & Nicolaysen K (1997). Hard training for 5 mo increases Na^+ - K^+ pump concentration in skeletal muscle of cross-country skiers. *Am J Physiol* **272**, 1417–1424.
- Faria EW, Parker DL & Faria IE (2005). The science of cycling: physiology and training – part 1. *Sports Med* **35**, 285–312.
- Fitts RH (2006). The muscular system: fatigue processes. In *ACSM's Advanced Exercise Physiology*, ed. Tipton CM, pp. 178–197. Lippincott Williams & Wilkins.
- Flagg TP, Enkvetchakul D, Koster JC & Nichols CG (2010). Muscle KATP channels: recent insights to energy sensing and myoprotection. *Physiol Rev* **90**, 799–829.
- Flatman JA & Clausen T (1979). Combined effects of adrenaline and insulin on active electrogenic Na^+ - K^+ transport in rat soleus muscle. *Nature* **281**, 580–581.
- Gaitanos GC, Williams C, Boobis LH & Brooks S (1993). Human muscle metabolism during intermittent maximal exercise. *J Appl Physiol* **75**, 712–719.
- Geers C & Gros G (2000). Carbon dioxide transport and carbonic anhydrase in blood and muscle. *Physiol Rev* **80**, 681–715.
- Gibala MJ & Jones AM (2013). Physiological and performance adaptations to high-intensity interval training. *Nestle Nutr Inst Workshop Ser* **76**, 51–60.

- Gibala MJ, Little JP, van Essen M, Wilkin GP, Burgomaster KA, Safdar A, Raha S & Tarnopolsky MA (2006). Short-term sprint interval *versus* traditional endurance training: similar initial adaptations in human skeletal muscle and exercise performance. *J Physiol* **575**, 901–911.
- Gong B, Legault D, Miki T, Seino S & Renaud JM (2003). K_{ATP} channels depress force by reducing action potential amplitude in mouse EDL and soleus muscle. *Am J Physiol Cell Physiol* **285**, C1464–C1474.
- Gosmanov AR, Fan Z, Mi X, Schneider EG & Thomason DB (2004). ATP-sensitive potassium channels mediate hyperosmotic stimulation of NKCC in slow-twitch muscle. *Am J Physiol Cell Physiol* **286**, C586–C595.
- Gosmanov AR, Schneider EG & Thomason DB (2003). NKCC activity restores muscle water during hyperosmotic challenge independent of insulin, ERK, and p38 MAPK. *Am J Physiol Regul Integr Comp Physiol* **284**, R655–R665.
- Grassi B, Rossiter HB & Zoladz JA (2015). Skeletal muscle fatigue and decreased efficiency: two sides of the same coin? *Exerc Sport Sci Rev* **43**, 75–83.
- Green HJ, Burnett M, Kollias H, Ouyang J, Smith I & Tupling S (2011). Malleability of human skeletal muscle sarcoplasmic reticulum to short-term training. *Appl Physiol Nutr Metab* **36**, 904–912.
- Green HJ, Chin ER, Ball-Burnett M & Ranney D (1993). Increases in human skeletal muscle Na^+ - K^+ -ATPase concentration with short-term training. *Am J Physiol* **264**, 1538–1541.
- Green HJ, Grange F, Chin C, Goreham C & Ranney D (1998). Exercise-induced decreases in sarcoplasmic reticulum Ca^{2+} -ATPase activity attenuated by high-resistance training. *Acta Physiol Scand* **164**, 141–146.
- Green S, Langberg H, Skovgaard D, Bulow J & Kjaer M (2000). Interstitial and arterial-venous $[K^+]$ in human calf muscle during dynamic exercise: effect of ischaemia and relation to muscle pain. *J Physiol* **529**, 849–861.
- Gunnarsson TP, Christensen PM, Holse K, Christiansen D & Bangsbo J (2012). Effect of additional speed endurance training on performance and muscle adaptations. *Med Sci Sports Exerc* **44**, 1942–1948.
- Gunnarsson TP, Christensen PM, Thomassen M, Nielsen LR & Bangsbo J (2013). Effect of intensified training on muscle ion kinetics, fatigue development, and repeated short-term performance in endurance-trained cyclists. *Am J Physiol Regul Integr Comp Physiol* **305**, R811–R821.
- Han F, Bossuyt J, Despa S, Tucker AL & Bers DM (2006). Phospholemman phosphorylation mediates the protein kinase C-dependent effects on Na^+ / K^+ pump function in cardiac myocytes. *Circ Res* **99**, 1376–1383.
- Hansen AK, Clausen T & Nielsen OB (2005). Effects of lactic acid and catecholamines on contractility in fast-twitch muscles exposed to hyperkalemia. *Am J Physiol Cell Physiol* **289**, C104–C112.
- Harmer AR, Ruell PA, Hunter SK, McKenna MJ, Thom JM, Chisholm DJ & Flack JR (2014). Effects of type 1 diabetes, sprint training and sex on skeletal muscle sarcoplasmic reticulum Ca^{2+} uptake and Ca^{2+} -ATPase activity. *J Physiol* **592**, 523–535.
- Hellsten Y, Apple FS & Sjödin B (1996). Effect of sprint cycle training on activities of antioxidant enzymes in human skeletal muscle. *J Appl Physiol* **81**, 1484–1487.
- Hill CA, Harris RC, Kim HJ, Harris BD, Sale C, Boobis LH, Kim CK & Wise JA (2007). Influence of beta-alanine supplementation on skeletal muscle carnosine concentrations and high intensity cycling capacity. *Amino Acids* **32**, 225–233.
- Hill CA, Thompson MW, Ruell PA, Thom JM & White MJ (2001). Sarcoplasmic reticulum function and muscle contractile character following fatiguing exercise in humans. *J Physiol* **531**, 871–878.
- Hobson RM, Saunders B, Ball G, Harris RC & Sale C (2012). Effects of β -alanine supplementation on exercise performance: a meta-analysis. *Amino Acids* **43**, 25–37.
- Hostrup M, Kalsen A, Ørtenblad N, Juel C, Mørch K, Rzeppa S, Karlsson S, Backer V & Bangsbo J (2014). β_2 -Adrenergic stimulation enhances Ca^{2+} release and contractile properties of skeletal muscles, and counteracts exercise-induced reductions in Na^+ - K^+ -ATPase $\dot{V}_{max}$ in trained men. *J Physiol* **592**, 5445–5459.
- Houston ME & Thomson JA (1977). The response of endurance-adapted adults to intense anaerobic training. *Eur J Appl Physiol Occup Physiol* **36**, 207–213.
- Hultman E & Sahlin K (1980). Acid-base balance during exercise. *Exerc Sport Sci Rev* **8**, 41–128.
- Iaia FM & Bangsbo J (2010). Speed endurance training is a powerful stimulus for physiological adaptations and performance improvements of athletes. *Scand J Med Sci Sports* **20**, 11–23.
- Iaia FM, Hellsten Y, Nielsen JJ, Fernström M, Sahlin K & Bangsbo J (2009). Four weeks of speed endurance training reduces energy expenditure during exercise and maintains muscle oxidative capacity despite a reduction in training volume. *J Appl Physiol* **106**, 73–80.
- Iaia FM, Thomassen M, Kolding H, Gunnarsson T, Wendell J, Rostgaard T, Nordsborg N, Krstrup P, Nybo L, Hellsten Y & Bangsbo J (2008). Reduced volume but increased training intensity elevates muscle Na^+ - K^+ pump $\alpha 1$ -subunit and NHE1 expression as well as short-term work capacity in humans. *Am J Physiol Regul Integr Comp Physiol* **294**, R966–R974.
- Jacobs RA, Flück D, Bonne TC, Bürgi S, Christensen PM, Toigo M & Lundby C (2013). Improvements in exercise performance with high-intensity interval training coincide with an increase in skeletal muscle mitochondrial content and function. *J Appl Physiol* **115**, 785–793.
- James JH, Wagner KR, King JK, Leffler RE, Upputuri RK, Balasubramaniam A, Friend LA, Shelly DA, Paul RJ & Fischer JE (1999). Stimulation of both aerobic glycolysis and Na^+ - K^+ -ATPase activity in skeletal muscle by epinephrine or amylin. *Am J Physiol* **277**, 176–186.
- Jensen L, Bangsbo J & Hellsten Y (2004). Effect of high intensity training on capillarization and presence of angiogenic factors in human skeletal muscle. *J Physiol* **557**, 571–582.
- Juel C & Halestrap AP (1999). Lactate transport in skeletal muscle – role and regulation of the monocarboxylate transporter. *J Physiol* **517**, 633–642.

- Juel C (1997). Lactate–proton cotransport in skeletal muscle. *Physiol Rev* **77**, 321–358.
- Juel C (1998a). Muscle pH regulation: role of training. *Acta Physiol Scand* **162**, 359–366.
- Juel C (1998b). Skeletal muscle Na^+/H^+ exchange in rats: pH dependency and the effect of training. *Acta Physiol Scand* **164**, 135–140.
- Juel C, Bangsbo J, Graham T & Saltin B (1990). Lactate and potassium fluxes from human skeletal muscle during and after intense, dynamic, knee extensor exercise. *Acta Physiol Scand* **140**, 147–159.
- Juel C, Hostrup M & Bangsbo J (2015). The effect of exercise and β_2 -adrenergic stimulation on glutathionylation and function of the Na,K-ATPase in human skeletal muscle. *Physiol Rep* **3**, pii: e12515.
- Juel C, Klarskov C, Nielsen JJ, Krstrup P, Mohr M & Bangsbo J (2004). Effect of high-intensity intermittent training on lactate and H^+ release from human skeletal muscle. *Am J Physiol Endocrinol Metab* **286**, E245–E251.
- Juel C, Nordsborg NB & Bangsbo J (2013). Exercise-induced increase in maximal *in vitro* Na-K-ATPase activity in human skeletal muscle. *Am J Physiol Regul Integr Comp Physiol* **304**, R1161–R1165.
- Juel C, Nordsborg NB & Bangsbo J (2014). Purinergic effects on Na,K-ATPase activity differ in rat and human skeletal muscle. *PLoS One* **9**, e91175.
- Juel C, Pilegaard H, Nielsen JJ & Bangsbo J (2000). Interstitial K^+ in human skeletal muscle during and after dynamic graded exercise determined by microdialysis. *Am J Physiol Regul Integr Comp Physiol* **278**, R400–R406.
- Kalsen A, Hostrup M, Backer V & Bangsbo J (2016). Effect of formoterol, a long-acting β_2 -adrenergic agonist, on muscle strength and power output, metabolism, and fatigue during maximal sprinting in men. *Am J Physiol Regul Integr Comp Physiol* **310**, R1312–R1321.
- Kemi OJ, Ellingsen O, Ceci M, Grimaldi S, Smith GL, Condorelli G & Wisløff U (2007). Aerobic interval training enhances cardiomyocyte contractility and Ca^{2+} cycling by phosphorylation of CaMKII and Thr-17 of phospholamban. *J Mol Cell Cardiol* **43**, 3354–3561.
- Klitgaard H & Clausen T (1989). Increased total concentration of Na-K pumps in vastus lateralis muscle of old trained human subjects. *J Appl Physiol* **67**, 2491–2494.
- Knicker AJ, Renshaw I, Oldham AR & Cairns SP (2011). Interactive processes link the multiple symptoms of fatigue in sport competition. *Sports Med* **41**, 307–328.
- Korzeniewski B & Zoladz JA (2002). Influence of rapid changes in cytosolic pH on oxidative phosphorylation in skeletal muscle: theoretical studies. *Biochem J* **365**, 249–258.
- Kristensen M & Juel C (2010). Potassium-transporting proteins in skeletal muscle: cellular location and fibre-type differences. *Acta Physiol (Oxf)* **198**, 105–123.
- Kristensen M, Hansen T & Juel C (2006). Membrane proteins involved in potassium shifts during muscle activity and fatigue. *Am J Physiol Regul Integr Comp Physiol* **290**, R766–R772.
- Lamb GD & Stephenson DG (2006). Point: lactic acid accumulation is an advantage during muscle activity. *J Appl Physiol* **100**, 1410–1412.
- Laursen PB (2010). Training for intense exercise performance: high-intensity or high-volume training? *Scand J Med Sci Sports* **20**, 1–10.
- Li JL, Wang XN, Fraser SF, Carey MF, Wrigley TV & McKenna MJ (2002). Effects of fatigue and training on sarcoplasmic reticulum Ca^{2+} regulation in human skeletal muscle. *J Appl Physiol* **92**, 912–922.
- Lindinger MI (2007). Combating muscle fatigue: extracellular lactic acidosis and catecholamines. *J Physiol* **581**, 419.
- Lindinger MI, Leung MJ & Hawke TJ (2013). Inward flux of lactate $^-$ through monocarboxylate transporters contributes to regulatory volume increase in mouse muscle fibres. *PLoS One* **8**, e84451.
- Lindinger MI, McKelvie RS & Heigenhauser GJ (1995). K^+ and Lac $^-$ distribution in humans during and after high-intensity exercise: role in muscle fatigue attenuation? *J Appl Physiol* **78**, 765–777.
- Lynch GS, McKenna MJ & Williams DA (1994). Sprint-training effects on some contractile properties of single skinned human muscle fibres. *Acta Physiol Scand* **152**, 295–306.
- MacIntosh BR & Shahi MR (2011). A peripheral governor regulates muscle contraction. *Appl Physiol Nutr Metab* **36**, 1–11.
- Madsen K, Franch J & Clausen T (1994). Effects of intensified endurance training on the concentration of Na,K-ATPase and Ca-ATPase in human skeletal muscle. *Acta Physiol Scand* **150**, 251–258.
- Majerczak J, Karasinski J & Zoladz JA (2008). Training induced decrease in oxygen cost of cycling is accompanied by down-regulation of SERCA expression in human vastus lateralis muscle. *J Physiol Pharmacol* **59**, 589–602.
- Mason MJ & Thomas RC (1988). A microelectrode study of the mechanisms of L-lactate entry into and release from frog sartorius muscle. *J Physiol* **400**, 459–479.
- Matar W, Nosek TM, Wong D & Renaud J (2000). Pinacidil suppresses contractility and preserves energy but glibenclamide has no effect during muscle fatigue. *Am J Physiol Cell Physiol* **278**, C404–C416.
- McKenna MJ, Bangsbo J & Renaud JM (2008). Muscle K^+ , Na^+ , and Cl disturbances and Na^+/K^+ pump inactivation: implications for fatigue. *J Appl Physiol* **104**, 288–295.
- McKenna MJ, Schmidt TA, Hargreaves M, Cameron L, Skinner SL & Kjeldsen K (1993). Sprint training increases human skeletal muscle $\text{Na}^+/\text{K}^+/\text{ATPase}$ concentration and improves K^+ regulation. *J Appl Physiol* **75**, 173–180.
- McKenzie DC, Parkhouse WS & Hearst WE (1982). Anaerobic performance characteristics of elite Canadian 800 meter runners. *Can J Appl Sport Sci* **7**, 158–160.
- Messonnier L, Kristensen M, Juel C & Denis C (2007). Importance of pH regulation and lactate/ H^+ transport capacity for work production during supramaximal exercise in humans. *J Appl Physiol* **102**, 1936–1944.
- Mohr M, Krstrup P, Nielsen JJ, Nybo L, Rasmussen MK, Juel C & Bangsbo J (2007). Effect of two different intense training regimens on skeletal muscle ion transport proteins and fatigue development. *Am J Physiol Regul Integr Comp Physiol* **292**, R1594–R1602.

- Munkvik M, Rehn TA, Slettaløkken G, Hasic A, Hallén J, Sjaastad I, Sejersted OM & Lunde PK (2010). Training effects on skeletal muscle calcium handling in human chronic heart failure. *Med Sci Sports Exerc* **42**, 847–855.
- Nevill ME, Boobis LH, Brooks S & Williams C (1989). Effect of training on muscle metabolism during treadmill sprinting. *J Appl Physiol* **67**, 2376–2382.
- Nielsen J, Cheng AJ, Ørtenblad N & Westerblad H (2014). Subcellular distribution of glycogen and decreased tetanic Ca^{2+} in fatigued single intact mouse muscle fibres. *J Physiol* **592**, 2003–2012.
- Nielsen JJ, Kristensen M, Hellsten Y, Bangsbo J & Juel C (2003). Localization and function of ATP-sensitive potassium channels in human skeletal muscle. *Am J Physiol Regul Integr Comp Physiol* **284**, R558–R563.
- Nielsen JJ, Mohr M, Klarskov C, Kristensen M, Krstrup P, Juel C & Bangsbo J (2004a). Effects of high-intensity intermittent training on potassium kinetics and performance in human skeletal muscle. *J Physiol* **554**, 857–870.
- Nielsen OB, de Paoli F & Overgaard K (2001). Protective effects of lactic acid on force production in rat skeletal muscle. *J Physiol* **536**, 161–166.
- Nielsen OB, Ørtenblad N, Lamb GD & Stephenson DG (2004b). Excitability of the T-tubular system in rat skeletal muscle: roles of K^{+} and Na^{+} gradients and Na^{+} – K^{+} pump activity. *J Physiol* **557**, 133–146.
- Nogueira L, Shiah AA, Gandra PG & Hogan MC (2013). Ca^{2+} -pumping impairment during repetitive fatiguing contractions in single myofibers: role of cross-bridge cycling. *Am J Physiol Regul Integr Comp Physiol* **305**, R118–R125.
- Nordsborg N, Mohr M, Pedersen LD, Nielsen JJ, Langberg H & Bangsbo J (2003). Muscle interstitial potassium kinetics during intense exhaustive exercise: effect of previous arm exercise. *Am J Physiol Regul Integr Comp Physiol* **285**, R143–R148.
- Nyberg M, Fiorenza M, Lund A, Christensen M, Rømer T, Piil P, Hostrup M, Christensen PM, Holbek S, Ravnholt T, Gunnarsson TP & Bangsbo J (2016). Adaptations to speed endurance training in highly trained soccer players. *Med Sci Sports Exerc* **48**, 1355–1364.
- Okamoto K, Wang W, Rounds J, Chambers EA & Jacobs DO (2001). ATP from glycolysis is required for normal sodium homeostasis in resting fast-twitch rodent skeletal muscle. *Am J Physiol Endocrinol Metab* **281**, E479–E488.
- Ørtenblad N, Lunde PK, Levin K, Andersen JL & Pedersen PK (2000). Enhanced sarcoplasmic reticulum Ca^{2+} release following intermittent sprint training. *Am J Physiol Regul Integr Comp Physiol* **279**, R152–R160.
- Ørtenblad N, Nielsen J, Saltin B & Holmberg HC (2011). Role of glycogen availability in sarcoplasmic reticulum Ca^{2+} kinetics in human skeletal muscle. *J Physiol* **589**, 711–725.
- Ørtenblad N, Westerblad H & Nielsen J (2013). Muscle glycogen stores and fatigue. *J Physiol* **591**, 4405–4413.
- Parkhouse WS & McKenzie DC (1984). Possible contribution of skeletal muscle buffers to enhanced anaerobic performance: a brief review. *Med Sci Sports Exerc* **16**, 328–338.
- Pedersen TH, de Paoli F & Nielsen OB (2005). Increased excitability of acidified skeletal muscle: role of chloride conductance. *J Gen Physiol* **125**, 237–246.
- Pedersen TH, de Paoli FV, Flatman JA & Nielsen OB (2009). Regulation of ClC-1 and K_{ATP} channels in action potential-firing fast-twitch muscle fibers. *J Gen Physiol* **134**, 309–322.
- Pedersen TH, Nielsen OB, Lamb GD & Stephenson DG (2004). Intracellular acidosis enhances the excitability of working muscle. *Science* **305**, 1144–1147.
- Pilegaard H, Bangsbo J, Richter EA & Juel C (1994). Lactate transport studied in sarcolemmal giant vesicles from human muscle biopsies: relation to training status. *J Appl Physiol* **77**, 1858–1862.
- Pilegaard H, Domino K, Noland T, Juel C, Hellsten Y, Halestrap AP & Bangsbo J (1999a). Effect of high-intensity exercise training on lactate/ H^{+} transport capacity in human skeletal muscle. *Am J Physiol* **276**, 255–261.
- Pilegaard H, Terzis G, Halestrap A & Juel C (1999b). Distribution of the lactate/ H^{+} transporter isoforms MCT1 and MCT4 in human skeletal muscle. *Am J Physiol* **276**, 843–848.
- Place N, Ivarsson N, Venckunas T, Neyroud D, Brazaitis M, Cheng AJ, Ochala J, Kamandulis S, Girard S, Volungevičius G, Paužas H, Mekideche A, Kayser B, Martinez-Redondo V, Ruas JL, Bruton J, Truffert A, Lanner JT, Skurvydas A & Westerblad H (2015). Ryanodine receptor fragmentation and sarcoplasmic reticulum Ca^{2+} leak after one session of high-intensity interval exercise. *Proc Natl Acad Sci USA* **112**, 15492–15497.
- Pollak KA, Swenson JD, Vanhaitsma TA, Hughen RW, Jo D, White AT, Light KC, Schweinhardt P, Amann M & Light AR (2014). Exogenously applied muscle metabolites synergistically evoke sensations of muscle fatigue and pain in human subjects. *Exp Physiol* **99**, 368–380.
- Puype J, Van Proeyen K, Raymackers JM, Deldicque L & Hespel P (2013). Sprint interval training in hypoxia stimulates glycolytic enzyme activity. *Med Sci Sports Exerc* **45**, 2166–2174.
- Rasmussen MK, Kristensen M & Juel C (2008). Exercise-induced regulation of phospholemman (FXD1) in rat skeletal muscle: implications for $\text{Na}^{+}/\text{K}^{+}$ -ATPase activity. *Acta Physiol (Oxf)* **194**, 67–79.
- Rose AJ, Kiens B & Richter EA (2006). Ca^{2+} -calmodulin-dependent protein kinase expression and signalling in skeletal muscle during exercise. *J Physiol* **574**, 889–903.
- Ross A & Leveritt M (2001). Long-term metabolic and skeletal muscle adaptations to short-sprint training: implications for sprint training and tapering. *Sports Med* **31**, 1063–1082.
- Sahlin K & Henriksson J (1984). Buffer capacity and lactate accumulation in skeletal muscle of trained and untrained men. *Acta Physiol Scand* **122**, 331–339.
- Selvin D & Renaud JM (2015). Changes in myoplasmic Ca^{2+} during fatigue differ between FDB fibers, between glibenclamide-exposed and Kir6.2/- fibers and are further modulated by verapamil. *Physiol Rep* **3**, pii: e12303.
- Shepley B, MacDougall JD, Cipriano N, Sutton JR, Tarnopolsky MA & Coates G (1992). Physiological effects of tapering in highly trained athletes. *J Appl Physiol* **72**, 706–711.

- Skovgaard C, Christensen PM, Larsen S, Andersen TR, Thomassen M & Bangsbo J (2014). Concurrent speed endurance and resistance training improves performance, running economy, and muscle NHE1 in moderately trained runners. *J Appl Physiol* **117**, 1097–1109.
- Sostaric SM, Skinner SL, Brown MJ, Sangkabutra T, Medved I, Medley T, Selig SE, Fairweather I, Rutar D & McKenna MJ (2006). Alkalosis increases muscle K^+ release, but lowers plasma $[K^+]$ and delays fatigue during dynamic forearm exercise. *J Physiol* **570**, 185–205.
- Spruce AE, Standen NB & Stanfield PR (1985). Voltage-dependent ATP-sensitive potassium channels of skeletal muscle membrane. *Nature* **316**, 736–738.
- Street D, Nielsen JJ, Bangsbo J & Juel C (2005). Metabolic alkalosis reduces exercise-induced acidosis and potassium accumulation in human skeletal muscle interstitium. *J Physiol* **566**, 481–489.
- Suko J, Maurer-Fogy I, Plank B, Bertel O, Wyskovsky W, Hohenegger M & Hellmann G (1993). Phosphorylation of serine 2843 in ryanodine receptor-calcium release channel of skeletal muscle by cAMP-, cGMP- and CaM-dependent protein kinase. *Biochim Biophys Acta* **1175**, 193–206.
- Swietach P, Youm JB, Saegusa N, Leem CH, Spitzer KW & Vaughan-Jones RD (2013). Coupled Ca^{2+}/H^+ transport by cytoplasmic buffers regulates local Ca^{2+} and H^+ ion signaling. *Proc Natl Acad Sci USA* **110**, 2064–2073.
- Thomas C, Bishop DJ, Lambert K, Mercier J & Brooks GA (2012). Effects of acute and chronic exercise on sarcolemmal MCT1 and MCT4 contents in human skeletal muscles: current status. *Am J Physiol Regul Integr Comp Physiol* **302**, R1–R14.
- Thomas C, Perrey S, Lambert K, Hugon G, Mornet D & Mercier J (2005). Monocarboxylate transporters, blood lactate removal after supramaximal exercise, and fatigue indexes in humans. *J Appl Physiol* **98**, 804–809.
- Thomassen M, Christensen PM, Gunnarsson TP, Nybo L & Bangsbo J (2010). Effect of 2-wk intensified training and inactivity on muscle Na^+-K^+ pump expression, phospholemman (FXD1) phosphorylation, and performance in soccer players. *J Appl Physiol* **108**, 898–905.
- Thomassen M, Gunnarsson TP, Christensen PM, Pavlovic D, Shattock MJ & Bangsbo J (2016). Intensive training and reduced volume increases muscle FXD1 expression and phosphorylation at rest and during exercise in athletes. *Am J Physiol Regul Integr Comp Physiol* **310**, R659–R669.
- Vorup J, Tybirk J, Gunnarsson TP, Ravnholt T, Dalsgaard S & Bangsbo J (2016). Effect of speed endurance and strength training on performance, running economy and muscular adaptations in endurance-trained runners. *Eur J Appl Physiol* **116**, 1331–1341.
- Walas H & Juel C (2012). Purinergic activation of rat skeletal muscle membranes increases V_{max} and Na^+ affinity of the Na,K -ATPase and phosphorylates phospholemman and $\alpha 1$ subunits. *Pflugers Arch* **463**, 319–326.
- Wallinga W, Meijer SL, Alberink MJ, Vliek M, Wienk ED & Ypey DL (1999). Modelling action potentials and membrane currents of mammalian skeletal muscle fibres in coherence with potassium concentration changes in the T-tubular system. *Eur Biophys J* **28**, 317–329.
- Walsh B, Howlett RA, Stary CM, Kindig CA & Hogan MC (2006). Measurement of activation energy and oxidative phosphorylation onset kinetics in isolated muscle fibers in the absence of cross-bridge cycling. *Am J Physiol Regul Integr Comp Physiol* **290**, R1707–R1713.
- Westerblad H & Allen DG (1993). The influence of intracellular pH on contraction, relaxation and $[Ca^{2+}]_i$ in intact single fibres from mouse muscle. *J Physiol* **466**, 611–628.
- Westerblad H & Allen DG (1996a). Mechanisms underlying changes of tetanic $[Ca^{2+}]_i$ and force in skeletal muscle. *Acta Physiol Scand* **156**, 407–416.
- Westerblad H & Allen DG (1996b). Slowing of relaxation and $[Ca^{2+}]_i$ during prolonged tetanic stimulation of single fibres from *Xenopus* skeletal muscle. *J Physiol* **492**, 723–736.
- Weston M, Taylor KL, Batterham AM & Hopkins WG (2014). Effects of low-volume high-intensity interval training (HIT) on fitness in adults: a meta-analysis of controlled and non-controlled trials. *Sports Med* **44**, 1005–1017.
- Xu H, Cui N, Yang Z, Wu J, Giwa LR, Abdulkadir L, Sharma P & Jiang C (2001). Direct activation of cloned K_{ATP} channels by intracellular acidosis. *J Biol Chem* **276**, 12898–12902.
- Zoladz JA, Grassi B, Majerczak J, Szkutnik Z, Korostyński M, Karasiński J, Kilarski W & Korzeniewski B (2013). *Exp Physiol* **98**, 883–898.

Additional information

Competing interests

None declared.